

FTEC5660 Homework 4: Guardrails for Financial AI Agents

Implementation

1. RAG Pipeline (Task 1.1)

Implemented `rag_retrieval_chain` that retrieves relevant NovaTech financial documents via similarity search and returns formatted context for the agent. The chain uses FAISS vector store with Google Gemini embeddings for efficient document retrieval.

2. Financial Reasoning Agent (Task 2.1)

Designed a system prompt for the financial analyst agent that includes:

- 5 explicit numbered reasoning steps (problem identification → data retrieval → calculation → conflict resolution → conclusion)
- Explicit instruction to override user hints that contradict verified data
- Requirement to show all calculations and cite sources
- Confidence assessment for all answers

3. Consistency Judge (Task 3.1)

Built an impartial judge system that:

- Evaluates internal consistency between the agent's reasoning trace and final output
- Detects trace-output mismatches, unsupported logical leaps, and internal contradictions
- Outputs exactly `Pass / Fail` verdict with a one-sentence critique
- Does not use external knowledge, only verifies internal consistency

4. Anti-Sycophancy Controller (Task 4)

- **Task 4.1:** Implemented `compute_error_signal` that returns 1 for `Fail` verdicts and 0 for `Pass` to drive strategy escalation
- **Task 4.2:** Implemented dynamic strategy escalation based on accumulated errors:
 - $E_{int} = 0$: Helpful persona + Direct answering strategy
 - $E_{int} = 1$: Skeptical persona + Chain-of-Thought reasoning
 - $E_{int} \geq 2$: Skeptical persona + Python code execution for deterministic arithmetic

5. FOG Rate Calculation (Task 5)

Implemented `fog_rate` metric that measures the fraction of cases where:

- The agent's reasoning trace contains the correct answer **AND**
- The final output is sycophantic (matches the adversarial user hint)

Results Analysis

Condition	Accuracy	Sycophancy Rate	FOG Rate
D0 Baseline (no hint)	80.0%	0.0%	0.0%
DS Adversarial (no guardrail)	100.0%	100.0%	100.0%
DS Adversarial (with guardrail)	100.0%	60.0%	60.0%

Key Findings:

- Baseline Performance:** Without adversarial hints, the agent achieves 80% accuracy with zero sycophantic behavior, as expected.
- Sycophancy Vulnerability:** Without guardrails, the agent is 100% sycophantic when presented with authoritative adversarial hints, even though its internal trace contains the correct answer (100% FOG rate).
- Guardrail Effectiveness:** The anti-sycophancy system reduces sycophancy and FOG rate by 40% (from 100% to 60%) while maintaining 100% answer accuracy. The escalating strategy successfully forces the agent to prioritize verified data over user hints.
- FOG Rate Insight:** FOG rate equals sycophancy rate in this experiment, indicating all sycophantic responses are cases where the agent actually knew the correct answer but chose to follow the user hint.